HARIN DULNETH

SOFTWARE ENGINEER INTERN

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SUMMARY

Computer Science undergraduate focused on building intelligent, production-grade systems at the intersection of software engineering and AI. Experienced in developing scalable backend services with FastAPI and .NET, and modern full-stack applications using React, Next.js, and TypeScript. Strong background in applied machine learning and LLM integration, working with PyTorch, TensorFlow, scikit-learn, Hugging Face, and high-performance inference APIs to deliver real-world predictive and generative solutions. Skilled in designing end-to-end ML pipelines, RESTful architectures, and data-driven systems backed by PostgreSQL, MongoDB, and Redis. Familiar with Docker and CI/CD workflows, with a focus on building reliable, maintainable, and high-performance applications. Passionate about translating complex AI capabilities into practical, user-centric products.

CORE COMPETENCIES

- **Programming Languages** - Python | C | C# | Java JavaScript | TypeScript
- **Backend** - FastAPI | Flask | Node.js | .NET Spring Boot | Express.js
- **Frontend** - React | Next.js | Tailwind
- **Machine Learning / AI** - TensorFlow | PyTorch Hugging Face | MLOps | LLM
- **Databases** - SQL | PostgreSQL | Redis MongoDB | Supabase
- **Other** - Git | Docker | CI/CD | Linux | Google Colab

PROJECTS

Trowd – Real-Time Transit Intelligence Platform | Project

Technical Stack – React (TypeScript), FastAPI, Node.js, PostgreSQL, Redis, Socket.IO, LightGBM, XGBoost

- Developed a full-stack real-time bus tracking and prediction system integrating real-time telemetry, historical segment data, and ML-based ETA forecasting through a fusion engine with confidence-weighted blending.
- Engineered a crowd-sourced arrival confirmation pipeline using ML-based arrival detection, GPS clustering, and temporal consistency checks, enhanced by a dynamic reporter reputation-based feature weighting system.
- Architected a self-improving feedback loop where real-world outcomes continuously retrain models, and implemented a low-latency real-time communication system using Socket.IO and Redis pub/sub for live updates on bus location, ETA, and occupancy.

CuraNex – AI Demand Forecasting & Inventory Optimization Platform | Project

Technical Stack – Python, React (TypeScript), TensorFlow, PyTorch, XGBoost, LightGBM

- Led development of an AI-driven demand forecasting platform to predict stock requirements, optimize supply chain efficiency, and detect inventory anomalies.
- Built a scalable ML pipeline integrating deep sequence models (TFT, NBEATS) and gradient boosting methods with dynamic recursive feature engineering.
- Designed an interactive analytics dashboard with advanced forecast visualization, anomaly alerts, and proactive inventory recommendation features to support decision-making.

AcademiTrend – Academic Analytics & Career Prediction Platform | Project

Technical Stack – Python, Flask, Scikit-learn, XGBoost, LightGBM, Prophet, ARIMA

- Led backend development of a data-driven academic analytics platform featuring multiple ML-powered prediction systems for course enrollment, pathway trends, and graduate career outcomes.
- Engineered advanced feature pipelines and developed 20+ RESTful APIs with Flask, incorporating automated model evaluation, hyperparameter tuning, trend analysis, and salary growth visualization to support data-driven insights.

Portfolio Website With AI Assistant | Live Website (https://portfolio-next-js-red-one.vercel.app/)

Technical Stack – Next.js 15, Tailwind CSS, Hugging Face, Upstash Vector DB, Groq (LLaMA 3.1), Redis

- Engineered a portfolio website with an AI assistant using Retrieval-Augmented Generation (RAG) to answer user queries based on indexed resume and GitHub data.
- Built a serverless AI pipeline integrating embeddings, vector search, and Redis-based session memory with automatic conversation summarization.
- Designed a highly interactive UI featuring WebGL shaders, Spline 3D elements, and dynamic visual effects for an engaging user experience.

CodeMart – Digital Project Marketplace | Project

Technical Stack – React (TypeScript), ASP.NET Core, Entity Framework Core, PostgreSQL (Supabase), Docker, Stripe API

- Developed a full-stack digital marketplace for buying and selling software projects with secure authentication, payment integration, and role-based access control.
- Built a RESTful backend supporting project listings, order management, reviews, and transaction workflows.
- Designed a responsive and visually rich frontend using Three.js, GSAP animations, and modern UI components to deliver a premium user experience.

EDUCATION

Bsc. in Computer Science (Hons) — *2023 - Present*

University of Kelaniya, Sri Lanka
Specialize in Artificial Intelligence (Current GPA - 3.82)

G.C.E Advanced Level - 1A 2Bs — *2019 - 2022*

Physical Science Stream
Isipathana College, Colombo 05

G.C.E Ordinary Level - 8As 1B — *2008 - 2018*

Isipathana College, Colombo 05

CERTIFICATES

- Deep Learning Specialization | DeepLearning.AI | Credential ID: PA5O7P3AQ85C
- AI & ML Engineering - Professional Certificate | Microsoft | Credential ID: DZ6XR637YI5Q
- Microsoft Azure for AI and Machine Learning | Microsoft | Credential ID: PUCA2NJTQZG2
- AI/ML Engineer - Stage 2 | SLIIT Faculty of Computing | Credential ID: txqnjcawyu
- IBM Cloud Essentials | IBM | Credential ID: ededb5b52cb14d2e97a793e6a9d9ca6f

ACHIEVEMENTS

- **KelaniXtreme - 1st Place (IEEE Student Branch University of Kelaniya)**
- **AITHON 2026 - Top 30 Finalist & MVP Phase (Hemas Holdings PLC)**
- **Selected for Presentation (FCTSRS 2025) – "Nexus: Centralized Platform for Planning and Managing Projects"**
- **Paticipated in SpiritX - Inter University Development Competition (MoraSpirit)**
- **Participated in IX'25 - Designathon (IEEE Student Branch of IIT)**

EXTRA-CURRICULAR

- Member - IEEE Sri Lanka Section (2024/25)
- Member - Computer Science Student Association (2023 - Present)
- Member - Free and Open-Source Software Community (2024 - Present)
- Rugby - 2nd Runners up (Inter Faculty Tournament 2025)